

RHCSA White Test

1. Reset root password

- Password: `trootent` via specific boot procedure

2. Configure network and set the static hostname

- IP Address: `172.25.250.10`
- Netmask: `255.255.255.0`
- Gateway: `172.25.250.2`
- DNS/Nameserver: `8.8.8.8`
- Hostname: `servera.rhcsa..com`

3. Configure YUM repositories using the provided links

Note: For the sake of this test to function correctly, do not modify anything under `/etc/yum.repos.d`. Instead, create two files named `BaseOS.repo` and `AppStream.repo` under the `/root` directory.

- Repositories: `BaseOS` and `AppStream`
- name inside the file should be `BaseOS` and `AppStream`
- BaseOS URL: `http://content.example.com/rhel9.0/x86_64/dvd/BaseOS`
- AppStream URL: `http://content.example.com/rhel9.0/x86_64/dvd/AppStream`

4. Debug SELinux for a web server running on a non-standard port (82)

- Change `httpd` to listen on `82`
- Serve existing files in `/var/www/html`
- Enable web service to auto-start on boot
- **Hint:** `firewall`

5. Create user accounts with a supplementary group

- Group: `sysadms`
- Users:
 - `natasha` and `harry` in group `sysadms`
 - `sarah` with non-interactive shell, not in `sysadms`
- Password for all users: `trootent`

6. Configure a cron job for user natasha

- Schedule: 13:40 daily
- Command: logger "EX200 in progress"

7. Create a collaborative directory

- Path: /home/manager
- Group: sysadms
- Permissions: Full permissions for "sysadms" group members, no access for other users except root.
- Files should inherit group ownership

8. Configure NTP

- Configure your machine as an NTP client for africa.pool.ntp.org

9. Configure AutoFS

- Automount NFS home directories for ldapuserX at /home/guests/ldapuserX
- NFS Server: classroom.example.com (172.25.250.254)
- Ensure login works for users ldapuser1 to ldapuser20

Note: For the sake of this test to function correctly, Write your answers in a file named /root/AutoFS using the following format with an absolute path like (/home/user/dir/file):

First file to modify : <name of the file>

Content of the first file : <insert content here>

Second file to modify : <name of the file>

Content of the second file : <insert content here>

The Command you execute after all of this : <insert command here>

10. Create user alex with UID 3456

- Password: trootent

11. Locate all files owned by harry

- Copy it under directory : /root/harry-files

12. Search for string ich

- File: /usr/share/dict/words
- Save results to: /root/lines

13. Create a compressed archive

- Archive: `/root/backup.tar.bz2`
- Source: `/usr/local`
- Compression: `bzip2` (ensure `bzip2` is installed)
- Notes: Use `tar -cfz` for `gzip`, `tar -cfJ` for `xz`

14. Create a script file `script.sh` in `/bin`

- Find files in `/usr/local` sized 30K–50K
- Copy to `/root/d1`
- Set SGID on `/root/d1`

15. Create a container image

- Source: `http://registry.redhat.access.com`
- Registry: `registry.redhat.access.com`
- Username: your username of the developer account
- Password: your password
- Execute as user: `athena`

16. Create and run a PDF converter container

- Create a user named `athena` with password `trootent` and install necessary container tools.
- Create directories `/data/input` and `/data/output` with proper ownership and permissions for the `athena` user (full access).
- Download the Python script and Dockerfile from `https://raw.githubusercontent.com/sachinyadav3496/Text-To-PDF/master/` (`pdf_converter.py` and `Dockerfile`).
- Build a container image named `pdf` and run a container named `pdfconverter` with volume mappings from `/data/input` to `/data/input` and `/data/output` to `/data/output` inside the container.
- Create a test file `file.txt` in the input directory and verify the PDF conversion works.

17. Configure systemd service for the PDF converter container

- As the **athena** user, create a systemd user service for the existing **pdfconverter** container.
- The service should be named **container-pdfconverter.service** and must be enabled to start automatically.
- Configure user linger for the athena user to ensure the service runs even when the user is not logged in.
- Verify that the service is active and will restart automatically on system reboot.

18. Create swap partition

- Create a 512MB swap partition which take effect automatically at boot-start.

19. Create volume group and logical volumes

- Create a volume group called **vgfs** of 160MB created on the **/dev/sdb** disk.

The PE size for the volume group should be set at 16MB. Create 2 logical volumes called **ext4vol** and **xfsvol** of size 80MB each and initialise them with the Ext4 and XFS file system types.

Ensure that both file systems are persistently defined using their logical volume device filenames. Create mount points **/ext4vol** and **/xfsvol**, mount the file systems, and verify their availability and usage.

20. Extend logical volume

- Extend the **xfsvol** logical volume along with the file system it contains. The new size should be between 190M and 175M.

21. Configure system tuning

- Set the recommended tuned profile as **aws**

Good Luck